Cyclicality of Formal Agricultural Credit: Do Banks Respond Differently to Weather, Production and Price Shocks? Evidence from Maharashtra and Bihar

Oct 26, 2026 · @Kanishka vats

Abstract

Formal agricultural credit in Bihar and Maharashtra does not rise after a dry monsoon; in Bihar it falls two to four years later, through fewer borrowers at regional rural banks. The study matches scheduled commercial bank credit (RBI BSR-1, 2004–2026) to IMD gridded monsoon rainfall, foodgrain yields and wholesale crop prices. State time series of about 20 years are too short to decide whether credit is countercyclical or procyclical: none of nine weather, production or price effects survives a multiple-testing correction. A panel of 71 districts (2010–2026), with district and state-by-year fixed effects, compares districts of the same state in the same year. A 20-point monsoon shortfall is associated with −0.7% credit the following March (90% interval −1.7% to +0.4%), which excludes the 3–5% rise reported for 1993–2010 by De and Vij (2013). In Bihar, credit is 2–4% lower two to four years after a shortfall; the fall is in the number of accounts and sanctioned limits, not loan size, and it is concentrated in regional rural banks and rural branches. Placebo monsoons after the credit date show no effect. None of the relationships forecasts credit growth better than a naive trend, so the accompanying monitoring dashboard is descriptive, not predictive.

1. Introduction

When the monsoon fails, farmers need credit most just as their capacity to repay weakens. Whether banks extend credit through such years (countercyclical) or pull back (procyclical) decides whether formal finance cushions agricultural shocks or deepens them.

This paper asks whether formal agricultural credit responds differently to weather, production and price shocks in Maharashtra and Bihar. The two states differ sharply: Bihar's agriculture is rice and wheat with little irrigation outside the north-west and a banking system in which regional rural banks (RRBs) are large; Maharashtra's is coarse cereals, pulses, oilseeds and cotton, with credit dominated by commercial banks and a metropolitan centre (Mumbai) that books much indirect agricultural lending.

We use three definitions throughout. Credit is **countercyclical** if it is higher after worse conditions, **procyclical** if lower, and **inconclusive** if the estimate cannot be distinguished from zero. All shocks are signed so that larger values mean worse conditions, so a positive coefficient is countercyclical.

The paper makes three contributions:

1. It updates De and Vij's (2013) district-level test of bank responses to drought to 2010–2026 for two contrasting states, and finds the opposite sign.
2. It locates the response in time (an event study), in the bank system (bank groups, branch location) and in the margin of adjustment (borrowers versus loan size).
3. It tests out of sample whether these relationships can forecast credit, as an early-warning system would require, and finds they cannot.

2. Related literature

Three strands of work frame the question: how rainfall shocks travel through the rural economy, how banks respond to them, and what drives formal agricultural credit in India.

Rainfall shocks and the rural economy. Brey and Hertweck (2023) show that monsoon shocks in India act asymmetrically: yields fall sharply after droughts while excess rain matters little. A widespread drought's short-lived yield loss leads to a persistent fall in wages and rise in food prices lasting up to five years. Shocks therefore leave a multi-year trail in household incomes, which is the horizon over which repayment capacity and credit demand can move. This motivates our lags of up to four years.

Bank responses to rural demand shocks. De and Vij (2013) use rainfall as an external shock to credit demand across up to 640 districts in 1993–2010. Their design has district and year fixed effects, a falsification test using neighbouring districts, and an event study around drought years. They find that agricultural credit rises by about 3–5% after a drought; the increase is in average loan size rather than the number of accounts, and private banks respond more than public banks. We adopt their timing (monsoon of year t−1 against credit at end-March t) and their identification, and use their estimate as a benchmark.

Determinants of formal agricultural credit. Mohan (2006) documents the growth of agricultural credit alongside weakening viability of rural credit institutions, notably cooperatives and RRBs, and argues for stronger institutions rather than more directed volume. Baria and Shamurailatpam (2022), using national data for 1991–92 to 2018–19, find that formal agricultural credit is driven mainly by demand-side factors (banking access, input use, rainfall instability). Supply is comparatively stable because priority-sector rules regulate it.

Gap. These studies are national, end before 2011, or treat credit as a single aggregate. None asks whether the response differs by shock type, by state, by lender type or over a multi-year horizon in the period of Kisan Credit Card expansion, interest subvention and state loan waivers. That is the gap this paper addresses for Bihar and Maharashtra.

3. Institutional background

Agricultural credit in India is directed credit. Priority-sector rules require banks to lend a share of their net credit to agriculture, and since 2015 a sub-target of 7–8% to small and marginal farmers. Crop loans run through the Kisan Credit Card (KCC), introduced in 1998–99 and revised in 2012 as an ATM-enabled card. Interest subvention since 2006–07 makes short-term crop loans up to Rs 3 lakh available at 7%, with extra relief for prompt repayment from 2009–10. After a declared natural calamity, RBI directions (consolidated in October 2018) allow banks to reschedule loans and extend fresh credit.

Policy events that can move credit independently of weather, and that the models control for or absorb:

Year	Scope	Event	Expected effect on credit outstanding
2006–07	All-India	Interest subvention on crop loans	Higher demand for short-term credit
2008	All-India	Agricultural Debt Waiver and Debt Relief Scheme (~Rs 52,000 crore)	Write-off lowers outstanding stock
2012–13	All-India	Revised, ATM-enabled KCC	Wider access
2015–16	All-India	Small and marginal farmer sub-target (7%, then 8%)	Shift towards small borrowers
2016	All-India	PMFBY crop insurance replaces NAIS (kharif 2016)	Insurance linked to crop loans
Nov 2016	All-India	Demonetisation of Rs 500/1000 notes	Cash shock in rabi season
Jun 2017	Maharashtra	Chhatrapati Shivaji Maharaj Shetkari Sanman Yojana (~Rs 34,000 crore)	Write-off; affects March 2018
2018	Bihar	Bihar Rajya Fasal Sahayata Yojana replaces PMFBY	State crop assistance

Year	Scope	Event	Expected effect on credit outstanding
Feb 2019	All-India	PM-KISAN, Rs 6,000 a year	Income support
Dec 2019	Maharashtra	Mahatma Jyotirao Phule Shetkari Karjmukti Yojana (up to Rs 2 lakh)	Write-off; affects March 2020
Mar 2020	All-India	RBI COVID-19 loan moratorium	Deferred repayments

The two states differ in ways that matter for the question. In March 2026, regional rural banks held 35% of Bihar's direct farm credit but 7% of Maharashtra's. Rural branches held 55% in Bihar and 31% in Maharashtra. Maharashtra books 13% of direct farm credit in metropolitan branches, mainly Mumbai. Bihar's crop area (2004–07) is 48% rice and 30% wheat. Maharashtra's is 34% coarse cereals, 18% pulses, 18% oilseeds and 15% cotton, so its farmers face more market-price risk. Maharashtra had two state loan waivers in the period; Bihar had none, which helps separate lending responses from write-offs.

4. Data and variable construction

All data are public and aggregate; every series was validated against an independent total before use.

Variable	Source	Unit and coverage	Construction
Agricultural credit	RBI DBIE, BSR-1 credit of scheduled commercial banks by district, bank group, population group and occupation	Rs crore outstanding at end-March; 2010–2026; 2.24 million rows	District totals reconcile exactly with RBI's state and national summaries
Longer state credit	RBI Handbook of Statistics on Indian States, Table 157	Rs crore; end-March 2004–2009	Agrees with DBIE within 0.003% on the 48 overlapping region-years
Monsoon rainfall	IMD 0.25° gridded daily rainfall	mm; 1971–2025, complete	June–September totals averaged over grid cells inside each state or district (Survey of India boundaries)

Variable	Source	Unit and coverage	Construction
Foodgrain output	RBI Handbook Tables 62, 71, 84	'000 t, '000 ha, kg/ha; 2004-05 to 2024-25	Yield checked against production / area (max gap 0.24%)
Crop prices	WPI, Office of the Economic Adviser	Index; 2004-05 to 2025-26	Spliced to 2011-12 = 100; deflated by the all-commodities WPI
Controls	RBI Handbook (agricultural GSVA, CPI); hand-coded policy events	Rs crore, %, dummies	—

Timing. Credit year t is the stock outstanding at end-March t. It is matched to the June–September monsoon of calendar year t−1, the agricultural year (t−1)–t and prices of financial year (t−1)–t, so every shock predates the credit it is compared with.

Shocks, defined before any estimation and signed so that larger is worse:

- *Weather*: monsoon shortfall = −(departure of June–September rainfall from its 1971–2020 mean, %); a drought flag at a departure of −20% or worse (IMD "deficient"); as robustness, rainfall below 75% or 70% of normal and a Standardised Precipitation Index (SPI) of −1 or less.
- *Production*: foodgrain yield as % below a region-specific linear trend.
- *Price*: area-weighted fall in real wholesale prices of the region's crops, with weights fixed at the 2004-05 to 2006-07 crop mix.

Credit measures. The main outcome is real (WPI-deflated) growth of total agricultural credit for state series and the log of direct (farmer) credit for districts. Credit is a stock, so it mixes new lending, renewals, restructuring and write-offs.

District panel. Of 77 credit districts, renamed districts (e.g. Aurangabad to Chhatrapati Sambhajinagar) were mapped to current boundaries; Thane and Palghar (split in 2014) were merged; Mumbai and Mumbai Suburban were excluded because their "agricultural" credit swings by thousands of crore a year and is not farm lending. This leaves 71 units (38 Bihar, 33 Maharashtra) and 1,207 district-years.

5. Empirical strategy

Two designs answer the question at two scales: state time series test all three shock types; a district panel gives the rainfall test enough power.

State time series (Bihar, Maharashtra, All-India, 2006–2026). For each region r, real credit growth g is regressed on the three adverse shocks, with a loan-waiver dummy in the headline model (M4):

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g_{rt} = \alpha_r + \beta_W\, W_{r,t-1} + \beta_P\, P_{r,t} + \beta_R\,
R_{r,t} + \gamma\, \mathrm{Waiver}_{rt} + \varepsilon_{rt}
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With about 20 observations, standard errors are Newey-West with one lag. P-values come from a restricted wild bootstrap (Rademacher weights, 9,999 draws), which is more reliable than t-tables at this size. Benjamini-Hochberg q-values control the false-discovery rate across the headline tests. Seven headline and eight robustness specifications were fixed in advance. Binary shocks with fewer than three events are not estimated: statewide averaging leaves Maharashtra with one drought year.

District panel (71 districts, 2010–2026). Following De and Vij (2013):

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100 \log C_{dt} = \alpha_d + \lambda_{s(d),t} + \sum_{k=1}^{3} \beta_k\,
\mathrm{Shortfall}_{d,t-k} + \varepsilon_{dt}
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District effects absorb permanent differences. State-by-year effects absorb everything common to a state in a year: waivers, prices, inflation, state policy and the state-wide monsoon. The coefficients are therefore identified only from rainfall differences between districts of the same state in the same year. Such differences are plausibly unrelated to bank behaviour. Standard errors are clustered by district, with wild-cluster-bootstrap p-values.

Pre-registered stress tests. Before estimation the following were written down: an event study with two placebo monsoons that fall after the credit date; standard errors clustered by both district and state-year; and a mechanism split of Bihar's response into accounts, average loan size, sanctioned limits, bank groups and branch location. The mechanism split's 24 tests form one Benjamini-Hochberg family.

Interpretation. A coefficient is labelled countercyclical or procyclical only when its wild-bootstrap p-value is below 0.10; otherwise it is inconclusive. Estimates are conditional associations: the design separates rainfall from state-wide confounders but cannot tell reduced supply from reduced demand.

6. Results

Credit does not rise after a dry monsoon in either state; in Bihar it falls two to four years later, through fewer borrowers at regional rural banks.

6.1 State time series: inconclusive

The state series cannot separate the three shocks. Effects are shown for an illustrative adverse shock: a 20-point monsoon shortfall, yield 10% below trend, or a 5% real crop-price fall (headline model M4, 2006–2025, n = 20).

Region	Shock	Effect on real credit growth (pp)	90% interval	Wild-bootstrap p	Reading
Bihar	Weather	−0.7	−3.3 to +1.9	0.65	Inconclusive
Bihar	Production	−2.5	−5.7 to +0.7	0.15	Inconclusive
Bihar	Price	−3.8	−7.0 to −0.6	0.08	Procyclical (q = 0.95)
Maharashtra	Weather	−2.0	−13.7 to +9.6	0.87	Inconclusive
Maharashtra	Production	+3.1	−3.5 to +9.6	0.54	Inconclusive
Maharashtra	Price	+1.9	−3.3 to +7.2	0.62	Inconclusive
All-India	Weather	+5.4	−1.7 to +12.6	0.30	Inconclusive
All-India	Production	−0.0	−10.2 to +10.1	1.00	Inconclusive
All-India	Price	−0.2	−2.6 to +2.2	0.87	Inconclusive

Only Bihar's price effect reaches 10% significance, and it does not survive the multiple-testing correction (q = 0.95) or most robustness checks (4 of 13). Maharashtra's intervals are wide because its credit series swings with metro-booked lending and the 2017 and 2019 waivers (real credit fell about 26% in the year to March 2020). These are absences of evidence, not evidence of no effect.

6.2 District panel: no immediate response

Comparing districts within the same state and year, a 20-point monsoon shortfall is associated with **−0.7%** direct credit the following March (90% interval −1.7% to +0.4%, p = 0.26). The interval excludes the 3–5% rise De and Vij (2013) found for 1993–2010. Separately, Bihar (−0.9%) and Maharashtra (−0.5%) are both inconclusive; their difference is 0.4 points (p = 0.77). A drought dummy (departure ⩽ −20%) gives +0.2% (p = 0.86), and deficits and excesses do not differ.

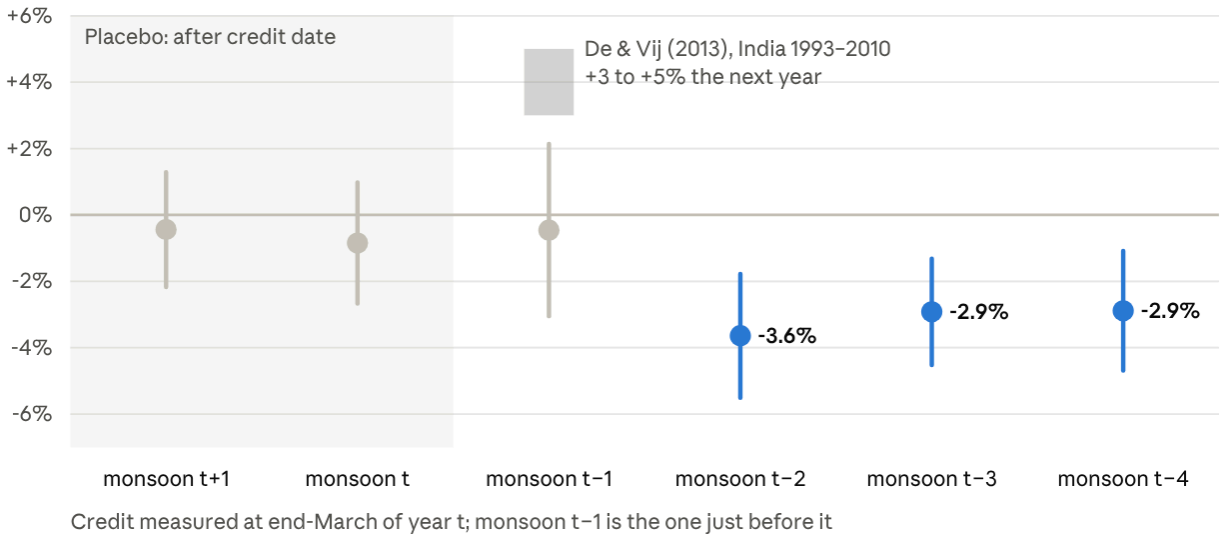
6.3 Bihar: credit falls two to four years later

In Bihar, a 20-point shortfall is followed by credit 3.9% lower two years on and 4.0% lower three years on (p ⩽ 0.001, q ≈ 0.01). The event study places the effect in time. The two monsoons that fall *after* the credit date show nothing, so drier districts were not already

on a lower path. The fall appears from the second year and persists into the fourth.

Bihar: credit is about 3% lower two to four years after a dry monsoon

% change in direct farm credit per 20-point monsoon shortfall; 38 districts, 2010–2024; lines are 90% intervals



Credit measured at end-March of year t; monsoon t−1 is the one just before it

Source: RBI DBIE BSR-1 credit, IMD gridded rainfall; district + year fixed effects, 38 Bihar districts, 2010–2024 (reports/tables/district_robustness.csv, spec E2)

The fall begins in the second year and does not reverse by the fourth; the earlier national benchmark is a rise of the opposite sign.

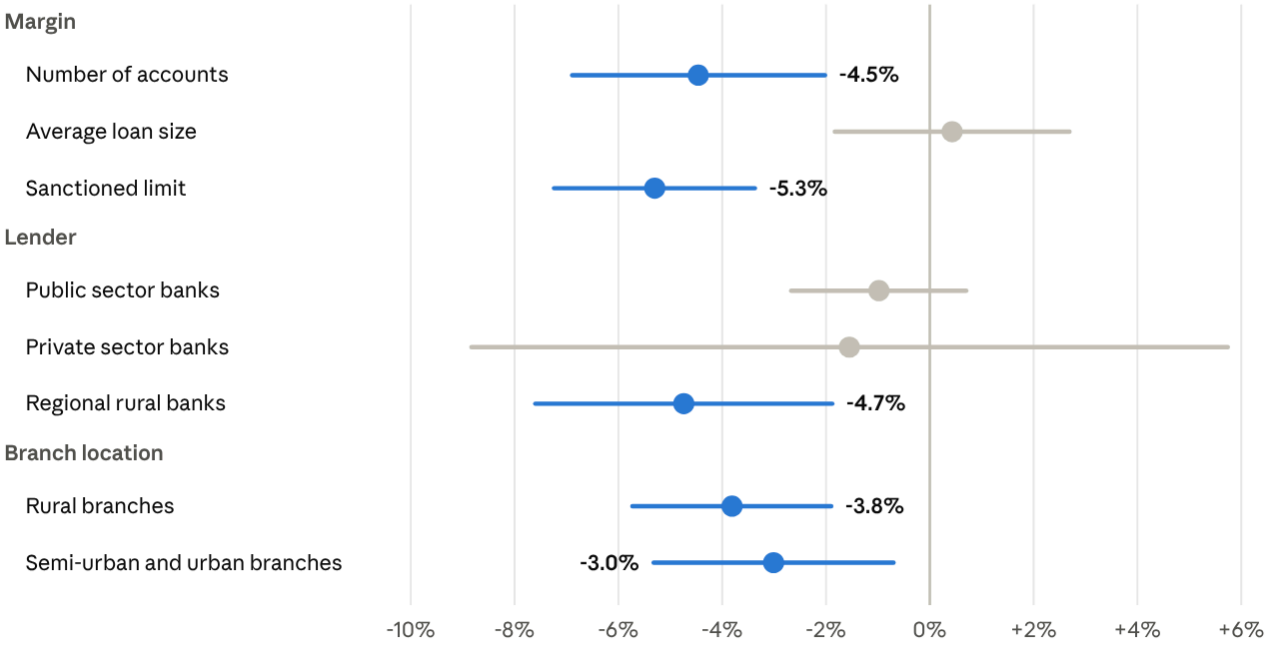
Maharashtra shows no delayed fall (+0.4% and +0.3%). Before its first waiver (credit years to 2017) it shows a small immediate rise of +1.7% ($p = 0.06$), which does not survive correction ($q = 0.23$).

6.4 Mechanism: fewer borrowers, at regional rural banks

The Bihar fall is on the **extensive margin**. Three years after a 20-point shortfall the number of direct-finance accounts is 4.5% lower ($q = 0.006$) and sanctioned limits 5.3% lower ($q = 0.005$), while the average loan is unchanged (+0.4%). The fall is concentrated in regional rural banks (−4.7%, $q = 0.02$) and rural branches (−3.8%, $q = 0.006$); public and private banks show no significant change. RRBs also lend a little more in the first year (+4.2%, $p = 0.045$, $q = 0.11$) before contracting. This is the reverse of De and Vij's national finding, where the response came through larger loans and private banks.

Bihar: three years after a dry monsoon, fewer farmers hold loans, mainly at RRBs

% change per 20-point shortfall in monsoon t−3; 38 districts, 2010–2026; lines are 90% intervals; blue = q < 0.10



Source: RBI DBIE BSR-1 credit by bank group and population group, IMD gridded rainfall; district + year fixed effects; q = Benjamini-Hochberg across 24 pre-registered tests (reports/tables/district_robustness.csv, specs K1–K8)

Average loan size does not move; the adjustment is in who holds a loan, at the lenders that serve small rural borrowers.

7. Forecasting and scenario analysis

None of these relationships forecasts credit better than a naive trend, so the scenario tool and early-warning monitor built on them are descriptive, not predictive.

Out-of-sample test. One-year-ahead forecasts of real credit growth were made by rolling origin: fit on all years up to T, forecast T+1, for T+1 = 2016–2025. The baseline is the naive forecast that credit grows at its past average. Error relative to that baseline (mean absolute error; below 1 is better):

Model	Bihar	Maharashtra	All-India
Naive: last year's growth	0.87	1.55	1.08
Autoregressive, one lag	0.89	1.10	0.87
Monsoon t−1 and t−2	1.04	1.08	1.11
Monsoon + yield + price	1.18	1.49	1.24

No model's gain over the naive forecast is significant (all $p \geq 0.10$). The shock models are worse out of sample than in it, the usual sign that a fitted relationship is too weak or unstable to forecast with. Mean absolute percentage errors run to 300–1,000% because growth is often near zero, so they are reported but not used. Holding out each of eight drought years in turn and fitting on all other years, the full shock model misses by 5.4 points on average against 6.1 for the naive forecast, and is closer in six of eight years; that is not significant (sign test $p = 0.29$) and, unlike the rolling test, it uses later years.

Can the Bihar lag rank districts? Using shortfalls two and three monsoons earlier to predict which Bihar districts will see below-average credit growth next year gives a mean rank correlation of 0.16 across 2017–2026 ($p = 0.10$) and no reduction in error. The lag is real in credit levels but small relative to year-to-year noise in district growth.

Scenario tool. The dashboard turns the estimates into scenarios in two ways. The district tool takes a monsoon departure and shows credit in each state relative to a normal monsoon over three years, in per cent and ₹ crore, with 90% bands: a 30% deficit implies Bihar credit about 3.5% (₹ 2,400 crore) lower three years on. The state tool combines monsoon, yield and price shocks for all three regions, with a prediction interval for next year's credit growth that is 30–60 points wide, and flags inputs outside the 2006–2025 range. Both are descriptive, given the forecast results above.

Early warning. A transparent rules-based monitor flags region-years as Normal, Watch or Elevated concern. Its thresholds are a monsoon departure of −20% or worse, yield more than one standard deviation below trend, a real price fall of more than one standard deviation, and real credit contraction. It marks the known droughts, Maharashtra's 2015 monsoon and Bihar's 2010 monsoon, as Elevated. In backtests it does not anticipate weaker credit: district drought flags are followed by below-median credit growth no more often than chance (lift 0.9–1.1). After the 2025 monsoon, 22 Bihar districts with deficits of 21–53% are on Watch, as is Satara in Maharashtra (−31%). A further 9 Bihar districts are flagged because of deficient monsoons in 2023–24, the lag found above. These flags call for local verification, not action.

8. Robustness and validation

The Bihar finding survives the pre-registered checks except district-specific trends, which shrink it by about 40%; the headline "no immediate rise" is robust throughout.

Check	Result for the Bihar lagged effect (t−2 / t−3, % per 20-point shortfall)	Verdict
Baseline (district + year effects, district clusters)	−3.9 / −4.0; p ⩽ 0.001	—
Placebo monsoons after the credit date	−0.4 (t+1), −0.8 (t); p = 0.66, 0.43	Passes: no pre-trend
Two-way clustering (district and state-year)	p = 0.003 / 0.023	Holds
District-specific linear trends	−2.3 / −2.3; p = 0.034 / 0.008; q = 0.16 / 0.06	Weakens by ~40%, still significant at t−3
Credit growth instead of level	−1.1 / −0.2 (both states); not significant	Effect is in levels, too small for growth
Maharashtra, same specification	+0.4 / +0.3; not significant	Bihar-specific

State-series robustness. Eight alternative specifications cover nominal instead of real credit, credit relative to All-India, direct credit only, non-metro credit, drought flag, standardised rainfall anomaly, excluding waiver years, and post-2010 only. None turns an inconclusive headline estimate into a robust one. Six further checks from the design plan, added after the headline results were known, give the same picture: drought defined as rainfall below 75% or 70% of normal (too few such years outside Bihar to estimate), SPI-based drought, and controls for non-agricultural credit growth or real GSDP growth leave every weather effect inconclusive. ADF and KPSS tests find all shock series stationary; real credit growth is stationary for Maharashtra and ambiguous for Bihar and All-India, as expected with 20 observations. One warning sign: in the state series a placebo test fails for Maharashtra (the following monsoon "predicts" earlier credit, p = 0.035). This is the expected rate of chance findings when about 40 tests are run, and a reason to read state-level significance cautiously.

Data validation. Credit totals reconcile exactly with RBI's published summaries; the long Handbook series matches DBIE to 0.003%. The all-India monsoon normal (859 mm) is close to IMD's long-period average (about 870 mm). Selected Handbook values and all policy-event dates were checked by hand against original publications; every event cites

an official source (for the Maharashtra waivers, the Government Resolutions of 28 June 2017 and 27 December 2019). The estimation code reproduces statsmodels' Newey-West and cluster standard errors exactly, and the full pipeline is covered by 34 automated tests.

9. Discussion, policy implications and limitations

Formal credit in these two states does not cushion monsoon shocks, and in Bihar it thins out after them, among the small rural borrowers RRBs serve.

Why might Bihar's credit fall with a lag? The pattern fits a repayment channel. A dry monsoon cuts incomes for several years (Brey and Hertweck 2023). Crop loans fall overdue, borrowers in default lose access to renewed KCC limits, and accounts close. Sanctioned limits fall along with accounts, consistent with banks not renewing them, and loan size does not change. RRBs, which lend most to small rural borrowers and carry thinner balance sheets, would feel this most. The data cannot confirm the channel: aggregate stocks do not show overdue loans, applications or rejections, so lower demand (farmers borrowing less after a loss) remains possible.

Why no immediate rise? De and Vij's 1993–2010 rise came through larger loans, plausibly restructuring and fresh calamity credit. Since then credit has been more heavily directed (KCC drives, priority-sector sub-targets, interest subvention) and, in Maharashtra, interrupted by waivers. Credit may now follow targets more than local conditions, in line with Baria and Shamurailatpam's (2022) finding that supply is stable and regulated.

Policy questions (hypotheses for the institutions that hold the data, not conclusions):

1. Do drought-hit Bihar borrowers become overdue and lose KCC renewals two to three years later? NABARD and the Bihar State Level Bankers' Committee could test this with account-level overdue data.
2. Are RRB balance sheets or capital limits binding after local droughts? NABARD supervises RRBs.
3. Are RBI natural-calamity relief measures (rescheduling, fresh loans) invoked and reaching borrowers in affected districts?
4. Would monitoring account numbers, not only credit volume, give earlier warning of exclusion after droughts?

Limitations.

- Credit is a stock: new lending, renewals, restructuring and write-offs cannot be separated.
- The mechanism (supply versus demand) is not identified.
- Crop prices are national; the states differ only through crop mix.

- State series are short; the district panel covers only rainfall shocks and 2010–2026.
- District units use current boundaries; Mumbai is excluded and Thane-Palghar merged.
- The Bihar effect shrinks by about 40% with district trends, so its size is uncertain even if its direction is robust.
- Results for two states need not generalise.

10. Conclusion

Formal agricultural credit in Bihar and Maharashtra is not countercyclical. Comparing districts of the same state in the same year, a dry monsoon brings no rise in credit the following year. The interval excludes the 3–5% rise documented for India before 2010. In Bihar, credit is lower two to four years after a shortfall because fewer farmers hold loans, mainly at regional rural banks and rural branches. That is procyclical exclusion rather than smaller loans. Maharashtra shows no clear response. State-level series of about 20 years are too short to say whether production or price shocks matter. None of these relationships forecasts credit growth, so monitoring tools built on them should point analysts to places worth checking, not predict outcomes. The most useful next step is account-level data on overdue loans and KCC renewals in drought-affected Bihar districts, to tell a repayment channel from reduced demand.

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Appendix: reproducibility

Every number in this paper is produced by a Python pipeline from the raw files above (Python 3.12, versions pinned). One command rebuilds the processed data, all models and an interactive dashboard: `python src/run_pipeline.py`. Specifications, the random seed (wild bootstrap, 9,999 draws) and library versions are saved with each model run. Sources, file checksums, a cleaning log and a data dictionary document every transformation. The accompanying dashboard (`app/dashboard.html`) is a single offline file with the results, a scenario tool and the district monitor.